Lab 1

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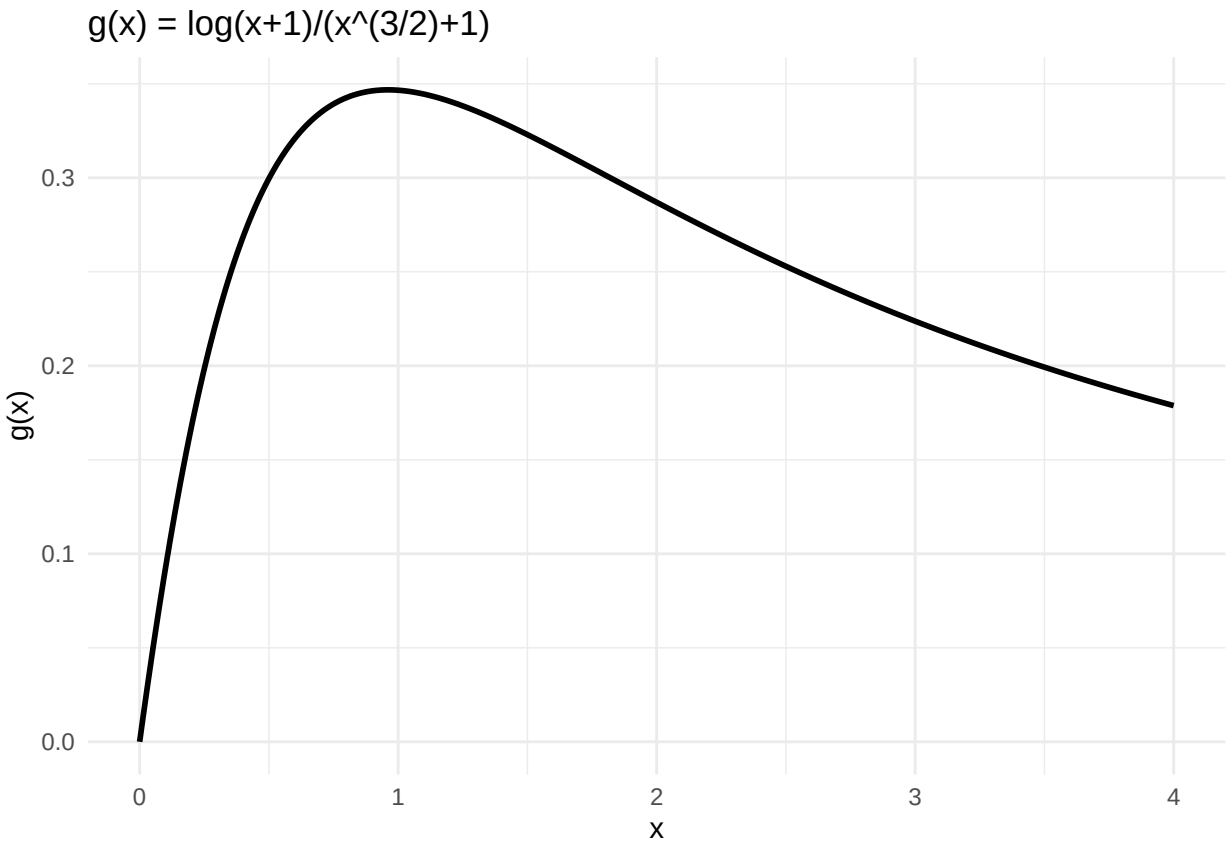
R Markdown

1.1. Function plot



The function maximum should be at around 1.

1.2. Derivative and plot

$g(x) = \dfrac{\log(x+1)}{x^{3/2}+1}$.

We set $u(x) = \log(x+1)$ and $v(x) = x^{3/2}+1$, so $g(x) = \dfrac{u(x)}{v(x)}$.

By the quotient rule,

$$g'(x) = \frac{u'(x)v(x) - u(x)v'(x)}{(v(x))^2}.$$

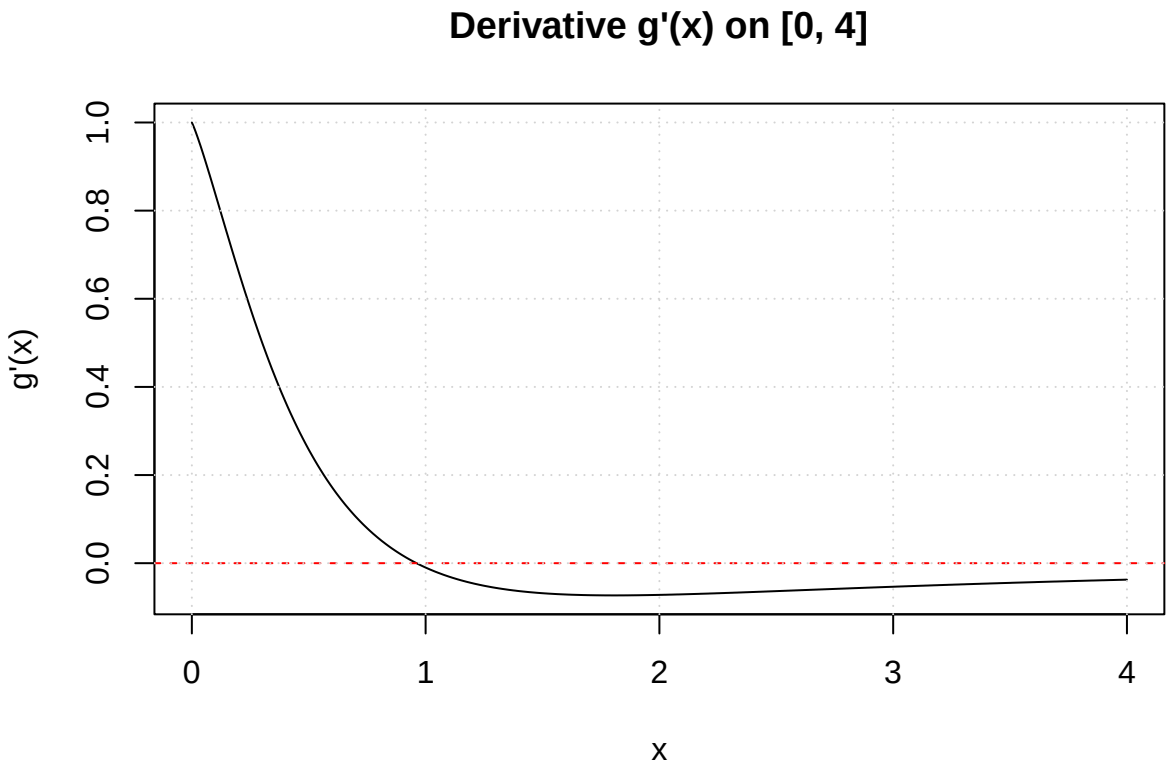
Then $u'(x) = \dfrac{1}{x+1}$ and $v'(x) = \dfrac{3}{2}x^{1/2}$.

Which gives:

$$g'(x) = \frac{\frac{1}{x+1}(x^{3/2}+1) - \log(x+1)\left(\frac{3}{2}x^{1/2}\right)}{(x^{3/2}+1)^2}.$$

Or equivalently,

$$g'(x) = \frac{(x^{3/2}+1) - \frac{3}{2}x^{1/2}(x+1)\log(x+1)}{(x+1)(x^{3/2}+1)^2}.$$



1.3. Bisection method implementation

Test results for $g(x)$ using the bisection method on the interval [0, 4] :

```
## $x_star
## [1] 0.9610603
##
## $g_at_x_star
## [1] 0.3467708
##
## $type
## [1] "local maximum (g' changes + to -)"
##
## $details
## $details$root
## [1] 0.9610603
```

```
##
## $details$f_root
## [1] -2.265823e-11
##
## $details$iter
## [1] 32
##
## $details$a
## [1] 0.9610603
##
## $details$b
## [1] 0.9610603
```

1.4. Secant method implementation

Test results for $g(x)$ using the secant method on the interval [0.2, 1] :

```
## $x_star
## [1] 0.9610603
##
## $g_at_x_star
## [1] 0.3467708
##
## $type
## [1] "local maximum (g' changes + to -)"
##
## $details
## $details$root
## [1] 0.9610603
##
## $details$f_root
## [1] 5.138024e-12
##
## $details$iter
## [1] 5
##
## $details$x0
## [1] 0.9610605
##
## $details$x1
## [1] 0.9610603
```

1.5. Comparison of the two methods for the task at hand

```
##       method start1 start2 status         root true_root_found iterations
## 1  bisection    0.0    0.5   fail           NA           FALSE         NA
## 2  bisection    0.0    1.0     ok 9.610603e-01            TRUE         30
## 4  bisection    0.0    1.5     ok 9.610603e-01            TRUE         29
## 7  bisection    0.0    2.0     ok 9.610603e-01            TRUE         31
## 11 bisection    0.0    2.5     ok 9.610603e-01            TRUE         32
## 16 bisection    0.0    3.0     ok 9.610603e-01            TRUE         30
## 22 bisection    0.0    3.5     ok 9.610603e-01            TRUE         29
## 29 bisection    0.0    4.0     ok 9.610603e-01            TRUE         32
## 3  bisection    0.5    1.0     ok 9.610603e-01            TRUE         29
## 5  bisection    0.5    1.5     ok 9.610603e-01            TRUE         30
```

```
## 8  bisection    0.5    2.0     ok 9.610603e-01            TRUE         31
## 12 bisection    0.5    2.5     ok 9.610603e-01            TRUE         31
## 17 bisection    0.5    3.0     ok 9.610603e-01            TRUE         31
## 23 bisection    0.5    3.5     ok 9.610603e-01            TRUE         32
## 30 bisection    0.5    4.0     ok 9.610603e-01            TRUE         32
## 6  bisection    1.0    1.5   fail           NA           FALSE         NA
## 9  bisection    1.0    2.0   fail           NA           FALSE         NA
## 13 bisection    1.0    2.5   fail           NA           FALSE         NA
## 18 bisection    1.0    3.0   fail           NA           FALSE         NA
## 24 bisection    1.0    3.5   fail           NA           FALSE         NA
## 31 bisection    1.0    4.0   fail           NA           FALSE         NA
## 10 bisection    1.5    2.0   fail           NA           FALSE         NA
## 14 bisection    1.5    2.5   fail           NA           FALSE         NA
## 19 bisection    1.5    3.0   fail           NA           FALSE         NA
## 25 bisection    1.5    3.5   fail           NA           FALSE         NA
## 32 bisection    1.5    4.0   fail           NA           FALSE         NA
## 15 bisection    2.0    2.5   fail           NA           FALSE         NA
## 20 bisection    2.0    3.0   fail           NA           FALSE         NA
## 26 bisection    2.0    3.5   fail           NA           FALSE         NA
## 33 bisection    2.0    4.0   fail           NA           FALSE         NA
## 21 bisection    2.5    3.0   fail           NA           FALSE         NA
## 27 bisection    2.5    3.5   fail           NA           FALSE         NA
## 34 bisection    2.5    4.0   fail           NA           FALSE         NA
## 28 bisection    3.0    3.5   fail           NA           FALSE         NA
## 35 bisection    3.0    4.0   fail           NA           FALSE         NA
## 36 bisection    3.5    4.0   fail           NA           FALSE         NA
## 37    secant    0.0    0.5     ok 9.610603e-01            TRUE          8
## 38    secant    0.0    1.0     ok 9.610603e-01            TRUE          5
## 40    secant    0.0    1.5     ok 9.610603e-01            TRUE         12
## 43    secant    0.0    2.0     ok 3.213281e+04           FALSE         34
## 47    secant    0.0    2.5     ok 3.366863e+04           FALSE         36
## 52    secant    0.0    3.0     ok 3.407846e+04           FALSE         36
## 58    secant    0.0    3.5     ok 3.601270e+04           FALSE         36
## 65    secant    0.0    4.0     ok 3.018935e+04           FALSE         35
## 39    secant    0.5    1.0     ok 9.610603e-01            TRUE          5
## 41    secant    0.5    1.5     ok 9.610603e-01            TRUE         10
## 44    secant    0.5    2.0     ok 3.417835e+04           FALSE         28
## 48    secant    0.5    2.5     ok 3.443344e+04           FALSE         36
## 53    secant    0.5    3.0     ok 3.366477e+04           FALSE         36
## 59    secant    0.5    3.5     ok 3.524994e+04           FALSE         36
## 66    secant    0.5    4.0     ok 2.947128e+04           FALSE         35
## 42    secant    1.0    1.5     ok 9.610603e-01            TRUE          7
## 45    secant    1.0    2.0     ok 9.610603e-01            TRUE          8
## 49    secant    1.0    2.5   fail           NA           FALSE         NA
## 54    secant    1.0    3.0     ok 3.456489e+04           FALSE         38
## 60    secant    1.0    3.5     ok 3.470520e+04           FALSE         38
## 67    secant    1.0    4.0   fail           NA           FALSE         NA
## 46    secant    1.5    2.0   fail           NA           FALSE         NA
## 50    secant    1.5    2.5     ok 3.321995e+04           FALSE         32
## 55    secant    1.5    3.0     ok 3.378095e+04           FALSE         34
## 61    secant    1.5    3.5     ok 3.116072e+04           FALSE         34
## 68    secant    1.5    4.0     ok 3.097144e+04           FALSE         34
## 51    secant    2.0    2.5     ok 3.531694e+04           FALSE         35
## 56    secant    2.0    3.0     ok 3.468978e+04           FALSE         35
```

```
## 62    secant    2.0    3.5     ok 3.543345e+04          FALSE         35
## 69    secant    2.0    4.0     ok 3.672040e+04          FALSE         35
## 57    secant    2.5    3.0     ok 3.403296e+04          FALSE         35
## 63    secant    2.5    3.5     ok 3.519499e+04          FALSE         35
## 70    secant    2.5    4.0     ok 3.668602e+04          FALSE         35
## 64    secant    3.0    3.5     ok 3.567116e+04          FALSE         35
## 71    secant    3.0    4.0     ok 2.921852e+04          FALSE         34
## 72    secant    3.5    4.0     ok 2.982723e+04          FALSE         34
```

The bisection method fails when the starting interval does not include the true root.

The secant method can succeed even when the starting interval does not include the true root and takes fewer iterations (up to 12 vs around 30 for the bisection method).

Using the update formula for the secant method

$$x_{n+1} = x_n - f(x_n)\,\frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})}.$$

which in our case is equivalent to

$$x_{n+1} = x_n - g'(x_n)\,\frac{x_n - x_{n-1}}{g'(x_n) - g'(x_{n-1})}.$$

The pattern in the table for the secant method (lots of roots around 10^4 or "fail") is consistent with secant taking a bad step and then wandering to a region where $g'(x)$ is very small/flat (we can observe it in the region of the $g'(x)$ graph in section 1.2 for in the x interval from 1.5 to 4 and we can see from the table that the algorithm fails in nearly all cases when one or both starting values are in this interval) or where the update becomes ill-conditioned (e.g. $g'(x_n) - g'(x_{n-1})$ tiny), which can blow up the update and lead to divergence or stagnation. This is a known secant pitfall: it can diverge when it encounters a near-zero slope / derivative-like quantity in the update.

```
## Algorithm converged:  TRUE
```

```
## Reason:  tolerance reached
```

```
## Number of iterations: 42
```

History - first 10 iterations:

```
##    iter        x0        x1          f_x0          f_x1         denom        x2
## 1     1  0.000000  2.000000  1.0000000000 -0.0719367338 -1.0719367338  1.865782
## 2     2  2.000000  1.865782 -0.0719367338 -0.0729762436 -0.0010395098 11.288245
## 3     3  1.865782 11.288245 -0.0729762436 -0.0062531156  0.0667231280 12.171294
## 4     4 11.288245 12.171294 -0.0062531156 -0.0053951572  0.0008579584 17.724226
## 5     5 12.171294 17.724226 -0.0053951572 -0.0025293110  0.0028658462 22.625080
## 6     6 17.724226 22.625080 -0.0025293110 -0.0015227392  0.0010065719 30.039079
## 7     7 22.625080 30.039079 -0.0015227392 -0.0008348680  0.0006878711 39.037437
## 8     8 30.039079 39.037437 -0.0008348680 -0.0004745677  0.0003603004 50.889574
## 9     9 39.037437 50.889574 -0.0004745677 -0.0002659422  0.0002086254 65.997914
## 10   10 50.889574 65.997914 -0.0002659422 -0.0001497870  0.0001161552 85.480754
##             f_x2
## 1  -7.297624e-02
## 2  -6.253116e-03
## 3  -5.395157e-03
## 4  -2.529311e-03
## 5  -1.522739e-03
## 6  -8.348680e-04
```

```
## 7  -4.745677e-04
## 8  -2.659422e-04
## 9  -1.497870e-04
## 10 -8.416287e-05
```

History - last 10 iterations:

```
##    iter        x0        x1          f_x0          f_x1        denom        x2
## 33   33  15533.31  19797.04 -4.481354e-10 -2.509786e-10 1.971568e-10  25224.73
## 34   34  19797.04  25224.73 -2.509786e-10 -1.405490e-10 1.104296e-10  32132.81
## 35   35  25224.73  32132.81 -1.405490e-10 -7.870170e-11 6.184731e-11  40923.45
## 36   36  32132.81  40923.45 -7.870170e-11 -4.406637e-11 3.463533e-11  52107.74
## 37   37  40923.45  52107.74 -4.406637e-11 -2.467168e-11 1.939469e-11  66335.10
## 38   38  52107.74  66335.10 -2.467168e-11 -1.381211e-11 1.085958e-11  84430.63
## 39   39  66335.10  84430.63 -1.381211e-11 -7.732002e-12 6.080103e-12 107442.52
## 40   40  84430.63 107442.52 -7.732002e-12 -4.328089e-12 3.403913e-12 136702.24
## 41   41 107442.52 136702.24 -4.328089e-12 -2.422556e-12 1.905534e-12 173900.90
## 42   42 136702.24 173900.90 -2.422556e-12 -1.355894e-12 1.066662e-12 221186.20
##             f_x2
## 33 -1.405490e-10
## 34 -7.870170e-11
## 35 -4.406637e-11
## 36 -2.467168e-11
## 37 -1.381211e-11
## 38 -7.732002e-12
## 39 -4.328089e-12
## 40 -2.422556e-12
## 41 -1.355894e-12
## 42 -7.588447e-13
```

We can see from the above output that with the starting values 0 and 2, due to the above-mentioned flatness of the $g'(x)$ graph, the secant algorithm overshoots the interval containing the root and then wanders further and further away from the interval containing the root as the numerator $x_n - x_{n-1}$ of the above equation becomes larger and larger while the denominator $g'(x_n) - g'(x_{n-1})$ becomes smaller and smaller.

The amount of programming effort is perhaps slightly higher for the secant method.

1.6. General comparison of the bisection and secant methods

For the task at hand, we would choose the bisection method despite it being slower than the secant method, we can determine the region containing the true root quite reliably, while the secant method appears to be rather sensitive to the choice of the starting values due to the flatness of the derivative function.

Generally, we prefer bisection when we have a bracketing interval $[a, b]$ with a sign change $f(a)f(b) < 0$, and we want a method that is guaranteed to converge under mild conditions (continuous f). We prefer secant when we have good starting values (close to the root), function evaluations are costly and we want fewer iterations than bisection.

Appendix

```r
#1.1
library(ggplot2)
g <- function(x) log(x + 1) / (x^(3/2) + 1)

df <- data.frame(x = seq(0, 4, length.out = 1000))
df$g <- g(df$x)
```

```r
ggplot(df, aes(x, g)) +
  geom_line(linewidth = 1) +
  labs(
    title = "g(x) = log(x+1)/(x^(3/2)+1)",
    x = "x", y = "g(x)"
  ) +
  theme_minimal()
#1.2
g_prime <- function(x) {
  ((x^(3/2) + 1)/(x + 1) - (3/2) * x^(1/2) * log(x + 1)) / (x^(3/2) + 1)^2
}

curve(g_prime(x), from = 0, to = 4, n = 1000,
      xlab = "x", ylab = "g'(x)",
      main = "Derivative g'(x) on [0, 4]")

abline(h = 0, lty = 2, col = "red")
grid()
#1.3

## Bisection method (root-finding) for a user-chosen interval [a, b]
bisect_root <- function(f, a, b, tol = 1e-10, max_iter = 200) {
  fa <- f(a); fb <- f(b)

  if (!is.finite(fa) || !is.finite(fb))
    stop("f(a) or f(b) is not finite.")

  if (fa == 0) return(list(root = a, f_root = fa, iter = 0, a = a, b = b))
  if (fb == 0) return(list(root = b, f_root = fb, iter = 0, a = a, b = b))

  ## Need a sign change for bisection
  if (fa * fb > 0)
    stop("No sign change on [a, b]. Choose an interval where f(a) and f(b) have opposite signs.")

  for (iter in 1:max_iter) {
    c <- (a + b) / 2
    fc <- f(c)

    if (!is.finite(fc))
      stop("f(midpoint) not finite; try a different interval.")

    if (abs(fc) < tol || (b - a) / 2 < tol)
      return(list(root = c, f_root = fc, iter = iter, a = a, b = b))

    if (fa * fc < 0) {
      b <- c; fb <- fc
    } else {
      a <- c; fa <- fc
    }
  }

  list(root = (a + b)/2, f_root = f((a + b)/2), iter = max_iter, a = a, b = b)
}
```

```r
## Find a critical point of g by solving g'(x)=0 on a user-selected interval
find_local_max_g <- function(a, b, tol = 1e-10, max_iter = 200) {
  out <- bisect_root(g_prime, a, b, tol = tol, max_iter = max_iter)
  x_star <- out$root

  ## First-derivative sign change test: + to - indicates local max
  eps <- min(1e-4, (b - a) / 100)
  left  <- g_prime(max(a, x_star - eps))
  right <- g_prime(min(b, x_star + eps))

  type <- if (is.finite(left) && is.finite(right) && left > 0 && right < 0) {
    "local maximum (g' changes + to -)"
  } else if (is.finite(left) && is.finite(right) && left < 0 && right > 0) {
    "local minimum (g' changes - to +)"
  } else {
    "critical point found; sign-change unclear on chosen interval"
  }

  list(x_star = x_star, g_at_x_star = g(x_star), type = type, details = out)
}

## Test
a = 0
b = 4
res <- find_local_max_g(a = a, b = b)
res
#1.4
## Secant method for root finding
secant_root <- function(f, x0, x1, tol = 1e-10, max_iter = 200) {
  f0 <- f(x0); f1 <- f(x1)

  if (!is.finite(f0) || !is.finite(f1))
    stop("f(x0) or f(x1) is not finite.")

  if (f0 == 0) return(list(root = x0, f_root = f0, iter = 0, x0 = x0, x1 = x1))
  if (f1 == 0) return(list(root = x1, f_root = f1, iter = 0, x0 = x0, x1 = x1))

  for (iter in 1:max_iter) {
    denom <- (f1 - f0)
    if (denom == 0) stop("Secant breakdown: f(x1) - f(x0) = 0.")

    ## x_{n+1} = x_n - f(x_n) (x_n - x_{n-1}) / (f(x_n) - f(x_{n-1}))
    x2 <- x1 - f1 * (x1 - x0) / denom

    if (!is.finite(x2)) stop("Non-finite iterate produced; try different starting values.")

    if (abs(x2 - x1) < tol)
      return(list(root = x2, f_root = f(x2), iter = iter, x0 = x0, x1 = x1))

    ## shift (x0,f0) <- (x1,f1), (x1,f1) <- (x2,f2)
    x0 <- x1; f0 <- f1
    x1 <- x2; f1 <- f(x1)
```

```r
    if (!is.finite(f1)) stop("Non-finite f(x) encountered; try different starting values.")
    if (abs(f1) < tol)  # optional early stop on function value
      return(list(root = x1, f_root = f1, iter = iter, x0 = x0, x1 = x1))
  }

  list(root = x1, f_root = f1, iter = max_iter, x0 = x0, x1 = x1)
}

## Find local max of g by applying secant to g'
find_local_max_g_secant <- function(x0, x1, tol = 1e-10, max_iter = 200) {
  out <- secant_root(g_prime, x0, x1, tol = tol, max_iter = max_iter)
  x_star <- out$root

  ## First-derivative sign-change check (heuristic)
  eps <- min(1e-4, abs(x1 - x0) / 100)
  left  <- g_prime(x_star - eps)
  right <- g_prime(x_star + eps)

  type <- if (is.finite(left) && is.finite(right) && left > 0 && right < 0) {
    "local maximum (g' changes + to -)"
  } else if (is.finite(left) && is.finite(right) && left < 0 && right > 0) {
    "local minimum (g' changes - to +)"
  } else {
    "critical point found; sign-change unclear from starting values"
  }

  list(x_star = x_star, g_at_x_star = g(x_star), type = type, details = out)
}

## Test
x0 = 0.2
x1 = 1
res <- find_local_max_g_secant(x0 = x0, x1 = x1)
res
## 1.5
# The "true" maximizer x* on [0,4]
# Find sign changes of g' on a fine grid, then bisect in the first bracket.
grid <- seq(0, 4, length.out = 2001)
vals <- g_prime(grid)

idx <- which(diff(sign(vals)) != 0)  # sign changes
if (length(idx) == 0) stop("No sign change found on [0,4]; cannot bracket a root.")

a_star <- grid[idx[1]]
b_star <- grid[idx[1] + 1]
true_out <- bisect_root(g_prime, a_star, b_star, tol = 1e-14, max_iter = 2000)
x_true <- true_out$root

## Bisection: test many [a,b]
bisect_grid <- expand.grid(
  a = seq(0, 3.5, by = 0.5),
  b = seq(0.5, 4.0, by = 0.5)
)
```

```r
bisect_grid <- subset(bisect_grid, a < b)

## Secant: test many (x0,x1) pairs
secant_grid <- expand.grid(
  x0 = seq(0.0, 3.5, by = 0.5),
  x1 = seq(0.5, 4.0, by = 0.5)
)
secant_grid <- subset(secant_grid, x0 < x1)

## Helper: run safely and return a row
run_one <- function(method, p1, p2, tol_root = 1e-4) {
  if (method == "bisection") {
    out <- tryCatch(bisect_root(g_prime, p1, p2), error = function(e) NULL)
    root <- if (is.null(out)) NA_real_ else out$root
    iter <- if (is.null(out)) NA_integer_ else out$iter
    status <- if (is.null(out)) "fail" else "ok"
    data.frame(method = method, start1 = p1, start2 = p2,
               status = status,
               root = root,
               true_root_found = isTRUE(!is.na(root) && abs(root - x_true) <= tol_root),
               iterations = iter)
  } else {
    out <- tryCatch(secant_root(g_prime, p1, p2), error = function(e) NULL)
    root <- if (is.null(out)) NA_real_ else out$root
    iter <- if (is.null(out)) NA_integer_ else out$iter
    status <- if (is.null(out)) "fail" else "ok"
    data.frame(method = method, start1 = p1, start2 = p2,
               status = status,
               root = root,
               true_root_found = isTRUE(!is.na(root) && abs(root - x_true) <= tol_root),
               iterations = iter)
  }
}

## Run and bind results
bisect_res <- do.call(rbind, lapply(seq_len(nrow(bisect_grid)), function(i)
  run_one("bisection", bisect_grid$a[i], bisect_grid$b[i])
))

secant_res <- do.call(rbind, lapply(seq_len(nrow(secant_grid)), function(i)
  run_one("secant", secant_grid$x0[i], secant_grid$x1[i])
))

results <- rbind(bisect_res, secant_res)

## Sort for readability
results <- results[order(results$method, results$start1, results$start2), ]

## Print summary table
results
# Investigate the cause of failure of the secant method
secant_trace <- function(f, x0, x1, tol = 1e-12, max_iter = 50) {
  hist <- vector("list", max_iter)
```

```r
  f0 <- f(x0); f1 <- f(x1)
  if (!is.finite(f0) || !is.finite(f1)) stop("Non-finite start.")

  for (iter in 1:max_iter) {
    denom <- f1 - f0

    x2 <- if (denom == 0) NA_real_ else x1 - f1 * (x1 - x0) / denom

    hist[[iter]] <- data.frame(
      iter = iter,
      x0 = x0, x1 = x1,
      f_x0 = f0, f_x1 = f1,
      denom = denom,
      x2 = x2,
      f_x2 = if (is.finite(x2)) f(x2) else NA_real_
    )

    if (!is.finite(x2))
      return(list(converged = FALSE, reason = "non-finite x2 (likely denom ~ 0 or overflow)",
                  iter = iter, history = do.call(rbind, hist[1:iter])))

    if (abs(x2 - x1) < tol || abs(f(x2)) < tol)
      return(list(converged = TRUE, reason = "tolerance reached",
                  iter = iter, root = x2, f_root = f(x2),
                  history = do.call(rbind, hist[1:iter])))

    ## shift
    x0 <- x1; f0 <- f1
    x1 <- x2; f1 <- f(x1)
    if (!is.finite(f1))
      return(list(converged = FALSE, reason = "non-finite f(x)",
                  iter = iter, history = do.call(rbind, hist[1:iter])))
  }

  list(converged = FALSE, reason = "max_iter reached",
       iter = max_iter, root = x1, f_root = f1,
       history = do.call(rbind, hist))
}

out <- secant_trace(g_prime, x0 = 0, x1 = 2, max_iter = 60)
cat("Algorithm converged: ", out$converged)
cat("Reason: ", out$reason)
cat("Number of iterations:", out$iter)
head(out$history, 10)
tail(out$history, 10)
```